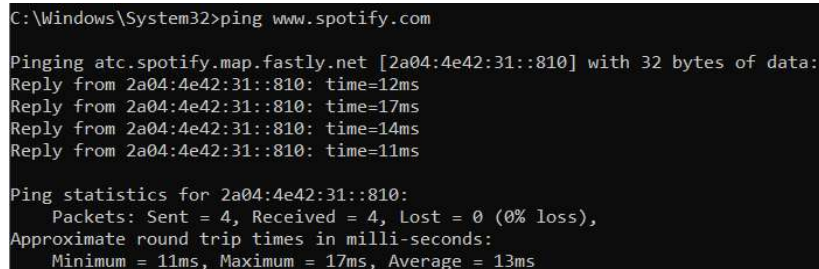


Using Ping and Traceroute

Q1. Open Command Prompt or Terminal and use the ping command to test connectivity to www.spotify.com. Note the number of packets sent, received, and the average response time.

Command used

```
ping www.spotify.com
```



```
C:\Windows\System32>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=12ms
Reply from 2a04:4e42:31::810: time=17ms
Reply from 2a04:4e42:31::810: time=14ms
Reply from 2a04:4e42:31::810: time=11ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 17ms, Average = 13ms
```

Figure 1: Output of ping www.spotify.com

Findings

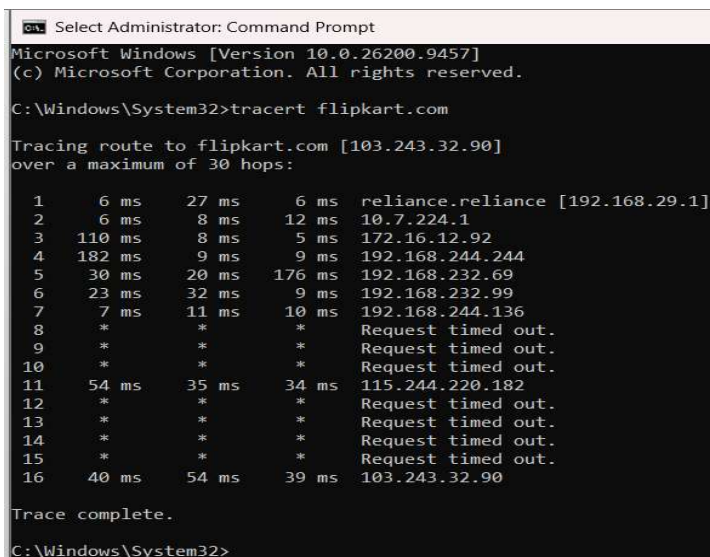
- **Resolved to:** atc.spotify.map.fastly.net (2a04:4e42:31::810)
- **Packets sent:** 4
- **Packets received:** 4 (0 lost, 0% loss)
- **Response time:** minimum 11 ms, maximum 17 ms, **average 13 ms**

All packets were returned with low response times, so connectivity to Spotify is working well.

Q2. Use the traceroute (tracert on Windows, traceroute on Linux/Mac) command to trace the path from your computer to www.flipkart.com. List the number of hops and identify the IP address of the first and last hop.

Command used

```
tracert flipkart.com
```



```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>tracert flipkart.com

Tracing route to flipkart.com [103.243.32.90]
over a maximum of 30 hops:

  0  6 ms   27 ms   6 ms   reliance.reliance [192.168.29.1]
  1  6 ms   8 ms   12 ms  10.7.224.1
  2 110 ms   8 ms   5 ms  172.16.12.92
  3 182 ms   9 ms   9 ms  192.168.244.244
  4  30 ms  20 ms  176 ms  192.168.232.69
  5  23 ms  32 ms   9 ms  192.168.232.99
  6   7 ms  11 ms  10 ms  192.168.244.136
  7  *      *      *      Request timed out.
  8  *      *      *      Request timed out.
  9  *      *      *      Request timed out.
 10  *      *      *      Request timed out.
 11 54 ms  35 ms  34 ms  115.244.220.182
 12  *      *      *      Request timed out.
 13  *      *      *      Request timed out.
 14  *      *      *      Request timed out.
 15  *      *      *      Request timed out.
 16 40 ms  54 ms  39 ms  103.243.32.90

Trace complete.

C:\Windows\System32>
```

Figure 2: Output of tracert [flipkart.com](https://www.flipkart.com)

Findings

- **Number of hops:** 16 (trace completed)
- **First hop:** 192.168.29.1 (reliance.reliance), which is the local router / default gateway
- **Last hop:** 103.243.32.90, which is the Flipkart destination server

Hops 8, 9, 10 and 12 to 15 show "Request timed out". The trace still reached the destination, so these routers are only not replying to traceroute requests (ICMP filtering). They are not faulty.

Q3. Simulate a connectivity issue by pinging a non-existent website (for example, ping www.fakezomato12345.com) and write down the error message you receive. Compare the output to a successful ping and describe the difference in your findings.

Command used

```
ping www.fake12345.com
```



```
C:\Windows\System32>ping www.fake12345.com
Ping request could not find host www.fake12345.com. Please check the name and try again.
```

Figure 3: Output of ping www.fake12345.com (non-existent website)

Error message received

```
Ping request could not find host www.fake12345.com. Please check the name and try again.
```

Comparison with a successful ping (Figure 1)

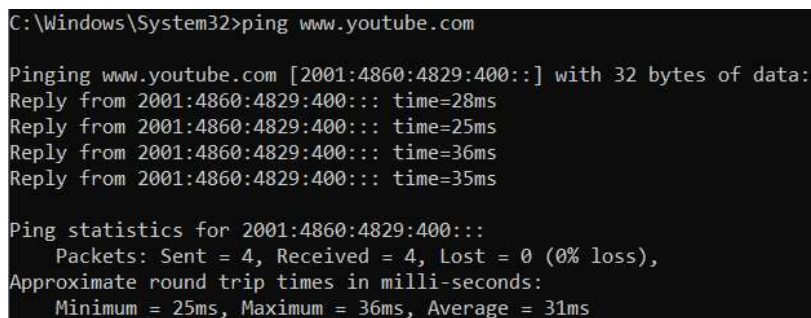
Point	Successful ping (spotify.com)	Failed ping (fake12345.com)
Name resolution	DNS resolved the name to an IP address	DNS could not find the host, so no IP address was found
Packets	4 sent, 4 received	No packets were sent
Replies	Reply lines with response times (11 to 17 ms)	No replies, only an error message
Statistics	Shows 0% loss and average time (13 ms)	No statistics shown

The failure happened at the name lookup (DNS) stage, before any data was sent. This is different from a "Request timed out" error, where the name resolves but the destination does not reply.

Q4. Imagine you are unable to access www.youtube.com from your system. Use both ping and traceroute commands to diagnose the issue and write a short report describing where the connection fails and what the possible reasons could be.

Commands used

```
ping www.youtube.com and tracert www.youtube.com
```



```
C:\Windows\System32>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4829:400::] with 32 bytes of data:
Reply from 2001:4860:4829:400::: time=28ms
Reply from 2001:4860:4829:400::: time=25ms
Reply from 2001:4860:4829:400::: time=36ms
Reply from 2001:4860:4829:400::: time=35ms

Ping statistics for 2001:4860:4829:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 36ms, Average = 31ms
```

Figure 4: Output of ping www.youtube.com

```
C:\Windows\System32>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4828:400::]
over a maximum of 30 hops:

  1    5 ms     6 ms     26 ms    2405:201:200f:c1cf:4e93:a6ff:fe8d:56c6
  2    *        *        *        Request timed out.
  3   12 ms     8 ms     10 ms    2405:203:400:100:172:31:3:177
  4   19 ms    16 ms    11 ms    2405:200:801:b00::ce4
  5   10 ms    15 ms    11 ms    2405:200:809:3168:61::5
  6    *        *        *        Request timed out.
  7  200 ms    27 ms    202 ms    2405:200:801:b00::dfc
  8    *        *        *        Request timed out.
  9   23 ms    74 ms    120 ms    2001:4860:4828:400::

Trace complete.
```

Figure 5: Output of tracert www.youtube.com

Report

- **Ping result:** 4 packets sent, 4 received, 0% loss, average 31 ms (minimum 25 ms, maximum 36 ms). The name resolved correctly and YouTube replied.
- **Traceroute result:** the trace reached the destination (2001:4860:4828:400::) in 9 hops and completed. Hops 2, 6 and 8 timed out, but the next hops replied, so those routers are just not answering traceroute requests.
- **Where the connection fails:** in this test it does not fail. Every step from the local router to YouTube worked, so the internet connection, DNS and route are all healthy.

What results mean if YouTube were not reachable

- **Ping says "could not find host":** the problem is DNS.
- **Traceroute stops at hop 1:** the problem is the local router or WiFi connection.
- **Traceroute stops at a hop inside the ISP network:** the problem is with the ISP or its routing.
- **Traceroute reaches the last few hops, then only "Request timed out":** the destination or a firewall in between is blocking the traffic.

Possible reasons for not being able to access YouTube

- DNS server problem or wrong DNS settings
- Firewall, parental control or content filter blocking the site
- Proxy, VPN or hosts file entry on the computer
- Local router or ISP outage
- IPv6 connectivity problem (the results above used IPv6 addresses)